

```

import java.util.ArrayList;
import java.util.InputMismatchException;
import java.util.Scanner;

public class ZooApp {
    public static void main(String[] args) {
        Zoo zoo = new Zoo();
        //creating my zoo object
        //loading my zoo function
        //creating my scanner
        Scanner sc = new Scanner(System.in);
        //creating a blank variable choice
        int choice;
        //creating a d while loop to ensure that the code runs correctly
        try {
            do {
                zoo.loadZooDetails();
                System.out.println("""
                    +===== ULSTER ZOO MENU =====+
                    1.| Add Animal
                    2 | Remove Animal
                    3 | Modify Animal
                    4 | Search Animal
                    5 | View all animals
                    6 | Daily care
                    7 | Zoo Report
                    8 | Movement
                    9 | Print Zoo Details
                    10| Update Zoo Details
                    0 | Exit
                    """);

                choice = sc.nextInt();
                System.out.println(""); //creating new line for input.
                //getting choice
                //taking the choice.

                if (choice == 1) {
                    zoo.loadAnimals();
                    addAnimalMenu(zoo, sc);
                    zoo.saveAnimals();
                }
            } while (choice != 0);
        } catch (InputMismatchException e) {
            System.out.println("Invalid input. Please enter a number between 0 and 10.");
        }
    }
}

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} //end of if add animal

else if (choice == 2) {
    zoo.loadAnimals();
    sc.nextLine();
    System.out.print("Enter name: ");
    String name = sc.nextLine();
    zoo.removeAnimal(name);
    zoo.saveAnimals();
} //end of else if remove animal

else if (choice == 3) {
    sc.nextLine();
    zoo.loadAnimals();
    modifyAnimal(zoo, sc);
    zoo.saveAnimals();
} //end of else if modify animal

else if (choice == 4) {
    zoo.loadAnimals();
    System.out.println("""
        Search with:
        1. Name
        2. Colour
        """);
    int searchChoice = sc.nextInt();
    sc.nextLine();
    if (searchChoice == 1) {
        System.out.print("Enter name: ");
        //asking the user to enter a name
        String name = sc.nextLine();
        //getting the name
        Animal found = zoo.searchByName(name);
        //searching by name with the function.

        if (found != null) {
            System.out.println(found.makeSound());
        } //end of if
    } //end of choice 1 if

    else if (searchChoice == 2) {

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System.out.print("Enter colour: ");
String colour = sc.nextLine();

ArrayList<Animal> results = zoo.searchByColor(colour);

if (results.isEmpty()) {
    System.out.println("No animals found with colour: " + colour);
} //end of first if
else {
    System.out.println(results.size() + " animal(s) found:");
    for (Animal a : results) {
        System.out.println(a.getName() + " (" + a.getClass().getSimpleName() +
"");

        System.out.println(a.makeSound()); //calling the make sound
        System.out.println();

        //get the user to type a color, call search method in zoo,
        //loops through animal list, made sure to ignore the case,
        //appends any matching animal colour to results
        //returns the matches.

        //get name, calls the get animal name, get class, get simple name =
shark, tiger
        // or parrot
    } //end of for
} //end of else
} //end else if of choice 2

else {
    System.out.println("Invalid search option.");
}
} //end of else if search animal 4

else if (choice == 5) {
    zoo.loadAnimals();
    zoo.listAnimals();
} //end of else if view all animals

else if (choice == 6) {
    zoo.loadAnimals();

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        zoo.dailyCare();
        System.out.println("If animal does not have specific care need, carry out\n" +
            " default caregiver needs, replace water, check bedding, clean\n" +
            " enclosure check health of animal and give medication if needed.\n" +
            " Contact head of medicated animal control if unsure.");
    } //end of else if Daily care

    else if (choice == 7) {
        zoo.loadAnimals();
        zoo.report();
    } //end of else if Zoo report

    else if (choice == 8) {
        zoo.loadAnimals();
        zoo.AnimalMovement();
    } //end of if add animal

    else if (choice == 9){
        zoo.printZooDetails();
    }
    else if (choice == 10){
        zoo.loadZooDetails();
        zoo.getNewZooDetails();
        zoo.saveZooDetails();
    }

    else if (choice == 0) {
        zoo.saveZooDetails();
        System.out.println("Exiting...");

    } //end of else if exit
    else {
        System.out.println("Invalid option. Please try again.");
    } //end of else

} while (choice != 0); //continue if user does not wish to exit
zoo.saveAnimals();
//calling zoo animals to repeats the program
System.out.println("Saved. Goodbye.");
//printing goodbye
}

```

```

catch (InputMismatchException e){
    System.out.println("Please Enter a valid Character");
}
catch(Exception e) {
    System.out.println("Please Enter a valid Number");
}
} // end of function

```

```

private static void addAnimalMenu(Zoo zoo, Scanner sc) {

```

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    System.out.println("1. Tiger 2. Parrot 3. Shark");
    int animalType = sc.nextInt();
    sc.nextLine(); //gets the input.

```

```

    System.out.print("Name: ");
    String addName = sc.nextLine();
    //gets the name

```

```

    System.out.print("Age: ");
    int addAge = sc.nextInt();
    sc.nextLine();
    //gets the age

```

```

    System.out.print("Color: ");
    String addColor = sc.nextLine();
    //gets the color

```

```

    System.out.print("Weight: ");
    double addWeight = sc.nextDouble();
    sc.nextLine();
    //gets the weight

```

```

    if (animalType == 1) { //checks if user input matches animal
        System.out.print("ClawLength: ");
        int addClawLength = sc.nextInt();
        sc.nextLine();
        zoo.addAnimal(new Tiger(addName, addAge, addColor, addWeight,
addClawLength));
    } //end of if, Tiger

```

```

else if (animalType == 2) { //checks if user input matches animal
    System.out.print("featherCondition: ");
    String featherCondition = sc.nextLine();
    zoo.addAnimal(new Parrot(addName, addAge, addColor, addWeight,
featherCondition));
    } //end of else if, parrot
else if (animalType == 3) { //checks if user input matches animal
    System.out.print("teethHealth (Rate 1-20): ");
    int teethHealth = sc.nextInt();
    sc.nextLine();
    zoo.addAnimal(new Shark(addName, addAge, addColor, addWeight,
teethHealth));
    } //end of else if, Shark
else {
    System.out.println("The animal you have entered can not be found...");
    } //end of else, prints if no animal is found.
} //end of function.

```

```

private static void modifyAnimal(Zoo zoo, Scanner sc) {
    zoo.loadAnimals();

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    System.out.print("Enter name: ");

```

```

    String name = sc.nextLine().trim();

```

```

    Animal a = zoo.searchByName(name);

```

```

    if (a == null) {
        System.out.println("Animals can not be found");
        return;
    } //end of if

```

```

    //getting the name of the animal and allowing me to edit the values

```

```

    System.out.println("""
        What would you like to modify?
        1. Age
        2. Weight
        3. Colour
        4. Name

```

5. Special Characteristic

0. Exit

```
""");
```

```
int choice = sc.nextInt();
```

```
sc.nextLine(); //getting the input form the user to see what they want to edit.
```

```
if (choice == 1) {
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```
    System.out.print("New age: ");
```

```
    int newAge = sc.nextInt();
```

```
    sc.nextLine();
```

```
    if (newAge > -1 ){
```

```
        a.setAge(newAge);
```

```
        System.out.println("Age updated.");
```

```
    } //end of check for zero
```

```
    else{
```

```
        System.out.print("Can not accept anything below 0 as an age.");
```

```
        return;
```

```
    } //end of else
```

```
} //end of if age
```

```
else if (choice == 2) {
```

```
    System.out.print("New weight: ");
```

```
    double newWeight = sc.nextDouble();
```

```
    sc.nextLine();
```

```
    if (newWeight > 0) {
```

```
        a.setWeight(newWeight);
```

```
        System.out.println("Weight updated.");
```

```
    } //end of if
```

```
    else {
```

```
        System.out.println("Weight must be greater than zero");
```

```
        return;
```

```
    } //end of else if weight check for 0
```

```
} // end of else if weight
```

```
else if (choice == 3) {
```

```
    System.out.print("New colour: ");
```

```
    String newColour = sc.nextLine();
```

```
    if (!newColour.isEmpty()) {
```

```
        a.setColor(newColour);
```

```

        System.out.println("Colour updated.");
    } //end of if for checking for nulls
    else if (newColour.isEmpty()){
        System.out.println(" Can not accept empty values ");
        return;
    } //end of else if for null
} //end of else if colour

else if (choice == 4) {
    System.out.print("New name: ");
    String newName = sc.nextLine(); //had to add this to stop
    //glitch with test, it was because it was not registering
    if (!newName.isEmpty()){
        a.setName(newName);
        System.out.println("Name updated.");
    } //end of if for check null
    else if (newName.isEmpty()){
        System.out.println("Can not accept empty values.");
        return;
    } //end of else if
} //end of else if name

else if (choice == 5) {
    //claw
    if (a instanceof Tiger) {
        Tiger t = (Tiger) a;
        System.out.print("New claw length: ");
        int newClaws = sc.nextInt();
        sc.nextLine();
        if (newClaws > 0) {
            t.setClawLength(newClaws);
            System.out.println("Claw length updated.");
        } //end of check for zero claw length
        else{
            System.out.println("Claw Length mst be greater than zero.");
            return;
        }
    }
    } //end of if claw
    //feather
    else if (a instanceof Parrot) {
        Parrot p = (Parrot) a;

```

```

    System.out.print("New feather condition: ");
    String newFeathers = sc.nextLine();
    if (!newFeathers.isEmpty()) {
        p.setFeatherCondition(newFeathers);
        System.out.println("Feather condition updated.");
    } //end of check for not blank
    else if (newFeathers.isEmpty()){
        System.out.println("Can not accept null values");
        return;
    } //end of checking for blank
} //end of if feather

// Shark ⇒ teeth health
else if (a instanceof Shark) {
    Shark s = (Shark) a;
    System.out.print("New teeth health: ");
    int newTeeth = sc.nextInt();
    sc.nextLine();
    if (newTeeth > 0 && newTeeth <= 20) {
        s.setTeethHealth(newTeeth);
        System.out.println("Teeth health updated.");
    } //end of check for zero.
    else if (newTeeth > 20){
        System.out.println("The teeth health must be between 1-20");
        return;
    } //end of check for above 15
    else {
        System.out.println("Teeth health can not be less than zero");
        return;
    } //end of teeth health
} //end of if teeth

else {
    System.out.println("You have selected an invalid animal");
} //end of else
} //end of else if extra attribute

else if (choice == 0) {
    System.out.println("Exiting...");
} //end of exit

```

```
    else {  
        System.out.println("The animal you have selected does not exist");  
    } //end of check if invalid  
} //end of modify animal  
  
} //end of class
```